

2.CAUTERY MACHINE

TECHNICAL SPECIFICATION

Item: Advanced Electrosurgical Generator / Energy Platform

Application: General Surgery, Laparoscopy, Gynaecology, Urology, ENT, Colorectal, Bariatric and other applicable surgical procedures.

Sl. No.	Parameter	Minimum Tender Specification
1	Type	Advanced microprocessor-controlled electrosurgical generator/energy platform
2	Technology	Suitable for monopolar and bipolar electrosurgery
3	Tissue Sensing	Real-time tissue impedance sensing technology for automatic adjustment of energy delivery according to tissue characteristics
4	Tissue Monitoring	High-frequency tissue monitoring and closed-loop energy control
5	Monopolar Cut	Shall provide Pure Cut and Blend Cut modes
6	Monopolar Coagulation	Shall provide Spray Coag, Fulguration, Soft Coag and other suitable coagulation modes
7	Bipolar Modes	Shall provide Precise, Standard and Macro Bipolar modes or equivalent
8	Advanced Bipolar	Shall support advanced bipolar vessel sealing instruments
9	Vessel Sealing	Integrated advanced vessel sealing technology with closed-loop control
10	Vessel Sealing Control	System shall continuously monitor tissue electrical characteristics and automatically regulate energy delivery
11	Automatic Power Control	Automatic power adjustment based on tissue characteristics shall be available
12	Bipolar Cable Compensation	System shall compensate for cable length/width variations to maintain consistent electrosurgical output
13	Touchscreen	Minimum 7-inch colour LCD touchscreen with user-friendly graphical interface
14	Display	Display shall provide selected mode, power setting, activation status, error/warning messages and relevant system information
15	Connector Ports	Minimum one standard monopolar connector, one advanced monopolar connector and one bipolar/advanced energy connector
16	Smart Connector	Smart connector system for automatic recognition/communication with compatible instruments, where applicable
17	Foot Control	Footswitch operation for activation/control of appropriate electrosurgical functions
18	Software	Software upgradeable through Ethernet or equivalent secure interface

Sl. No.	Parameter	Minimum Tender Specification
19	Output	Isolated electrosurgical output with appropriate patient safety monitoring
20	Safety	Shall incorporate patient safety features, fault detection and audible/visual activation and error alerts
21	Power Supply	220–240 VAC, 50–60 Hz
22	Input Power	Maximum approximately 950 VA at nominal line voltage or better
23	Cooling	Natural convection and fan-assisted cooling
24	Enclosure	Robust medical-grade enclosure suitable for operating-room use
25	Dimensions	Approximately 18 × 37 × 46 cm or compact equivalent
26	Weight	Approximately 10 kg or less
27	Mounting	Suitable for equipment cart, OR boom system or stable medical equipment table
28	Operating Temperature	Approximately 10–40°C
29	Operating Humidity	Suitable for hospital operating-room environmental conditions
30	Duty Cycle	Capable of continuous surgical use as per manufacturer's specified duty cycle
31	Applications	Suitable for open and minimally invasive surgical procedures
32	Instrument Compatibility	Shall support compatible monopolar, bipolar and advanced energy instruments
33	Vessel Sealing Instruments	Compatible with a comprehensive range of reusable/disposable advanced vessel-sealing instruments
34	Smoke Evacuation	Should be compatible with an integrated or external electrosurgical smoke evacuation system
35	Installation	Installation, commissioning and functional demonstration at hospital site shall be included
36	Training	Comprehensive user training for surgeons, OT nurses and biomedical engineering personnel
37	Warranty	Minimum 3 years comprehensive warranty from date of installation and commissioning
38	Service Support	Authorized service support with trained service engineers available in India
39	Preventive Maintenance	Preventive maintenance schedule and service support as per manufacturer's recommendations
40	Documentation	User manual, service manual, technical datasheet and electrical safety documentation shall be supplied
41	Regulatory Compliance	The equipment shall comply with applicable Indian regulatory requirements and relevant international medical electrical safety standards

Sl. No.	Parameter	Minimum Tender Specification
42	Demonstration	Bidder shall arrange product demonstration, if requested by the purchaser/technical evaluation committee

Minimum Required Accessories

The quoted system should be supplied as a **complete operational package**, including:

1. Advanced electrosurgical generator – **1 No.**
2. Compatible footswitch – **1 No.**
3. Patient return electrode monitoring system – integrated
4. Monopolar active electrode/pencil with compatible accessories
5. Monopolar connecting cable – suitable quantity
6. Bipolar connecting cable – suitable quantity
7. Compatible bipolar forceps
8. Advanced vessel-sealing instrument(s), as specified by the hospital
9. Patient return electrode plates – initial quantity as required
10. Equipment trolley/cart – **1 No.**, if required
11. All required power and connection cables
12. User manual – **1 set**
13. Service/technical manual – **1 set**
14. Complete installation and commissioning
15. On-site application training

Warranty & After-Sales Support

- **Comprehensive Warranty:** Minimum 3 years from the date of installation/commissioning.
- **Preventive Maintenance:** Minimum 4 preventive maintenance visits per year during the warranty period.
- **Spare Parts Support:** Availability and support for genuine spare parts for a minimum period of 10 years.
- **Backup Machine:** In the event of any equipment complaint or breakdown, a suitable backup machine shall be arranged within 12 hours until the original equipment is restored to operational condition.
- **AMC/CAMC Charges:** The AMC/CAMC rate as a percentage of the equipment purchase value, applicable after the warranty period, shall be finalized and frozen at the time of purchase and clearly mentioned in the Purchase Order.
- **Manufacturer Authorization & Warranty:** All authorization letters, warranty documents, and related commitments shall be issued directly by the original manufacturer/company, and not solely by the vendor/dealer.